

Figure A1. The feasible set of wheat-candy pairs in the example.

so a particular set of (possibly) nonnegative multipliers, according to complementary slackness. In this particular case, the picture won't cut down a lot on the algebra needed to find the solution, but in other applications, knowing how to draw this sort of figure and relate it to sets of binding constraints (and nonnegative multipliers) can be a substantial help. (Imagine, for example, that I added a third constraint on the amount of cholesterol that the consumer can take in. Then knowing which pairs of constraints can simultaneously bind can be a big help.)

Second, one thing we didn't do is compute the consumer's utility at the optimal solution given above. It is

$$3 \log(7.8) + 2 \log(2 + 1.9) = 8.88432431 \dots$$

Now let me suppose that this particular consumer, having been something of a success at his diet, is allowed by his doctor to increase his caloric intake by one calorie to 1,551 per day. If we go back and solve the problem, we find that the new optimal levels of wheat and candy are $c = 951/500 = 1.902$ and $w = 1951/250 = 7.804$. This, in turn, gives him utility $3 \log(7.804) + 2 \log(2 + 1.902) = 8.88688776 \dots$. So the additional utility that he gets from one extra calorie is $.00256345 \dots$, or, to the third significant decimal place, the value of the multiplier ν . An amazing coincidence? No — read on.

A1.3. Intuition

The intuition behind this recipe is most easily communicated in the special case where $k = 1$; that is, where the vector x is one-dimensional. Of course, this special case hides some of the subtleties. But if you see

why things work in this special case, then it shouldn't be too hard to imagine that it will work in general — the principles are exactly the same — and you should be able to consult a book on optimization and nonlinear programming to see this done more carefully and in greater generality.

Moreover, I'm going to make a number of assumptions that in total mean that the problem we're trying to solve is "well behaved." To be precise, I'm going to assume that the problem we're trying to solve is

$$\max f(x) \text{ subject to } g_1(x) \leq c_1 \text{ and } g_2(x) \leq c_2.$$

I'm going to assume throughout that the three functions given are differentiable everywhere, that the set of x satisfying the two constraints is a finite union of disjoint intervals, that there is no point x at which both constraints bind simultaneously, and that if $g_i(x) = c_i$ at some point x , then the derivative of g_i at this point is either strictly positive or strictly negative. (If you are reviewing this material after encountering it previously, you may recognize that I have just assumed the great-granddaddy of all constraint qualifications.)

A review of unconstrained optimization

But before we take on this problem in constrained optimization, we review unconstrained optimization. Suppose that the problem was to maximize $f(x)$ without any constraints at all. Then (this is the part you're supposed to know) you find the optimum by setting the derivative of f to zero. This is called the *first-order condition*, and it is a necessary condition for finding a maximum value of f . Why? Because if at a point x^0 the derivative of f is positive, then by increasing x a bit from x^0 you'll get a bigger value of f . And if the derivative of f is negative at x^0 , then f gets bigger if you decrease x a bit from x^0 .

You may also recall that finding all the points where $f' = 0$ generates all candidates for the optimum, but there is generally no guarantee that any solution of this FOC is even locally a maximum. These solutions, called *critical points*, could be minima. Or they could be points like the point 0 in the function x^3 , which are neither local minima nor maxima. To know that we have a local maximum, we check the second-order condition: If f is twice continuously differentiable and $f'' < 0$ at a point where $f' = 0$, then that point is a local maximum. (Why? If you know how to create Taylor's series expansions, you should be able to see why the condition $f'' < 0$ is sufficient for a local maximum. Note that $f'' \leq 0$ is necessary for this, but that if $f'' = 0$, we need to look at higher order derivatives, assuming f has them.)

And even if we have a local maximum, there is no guarantee in general that it is a global max. This is, however, guaranteed if f is concave. Then any local maximum is a global maximum. Why? Suppose the function f is concave. That is, $f(ax + (1 - a)y) \geq af(x) + (1 - a)f(y)$ for all x, y and $0 \leq a \leq 1$. Then if x isn't a global max, there is some value y with $f(y) > f(x)$. But then $f(ax + (1 - a)y) \geq af(x) + (1 - a)f(y) > f(x)$ for all a greater than zero and less than one. But this says that x isn't a local max, since for a arbitrarily close to (and less than) one, $ax + (1 - a)y$ is arbitrarily close to x . Moreover, if f is concave, any solution of $f' = 0$ is automatically a local and hence a global max. (If you never knew this before, just take my word for it.) So if f is concave, we know that satisfaction of the FOC is necessary and sufficient for a global maximum.

Drawing pictures

Now back to our simple example of constrained optimization: maximize $f(x)$ subject to $g_1(x) \leq c_1$ and $g_2(x) \leq c_2$. Recall that we assumed that the set of x satisfying the two constraints is a finite union of disjoint intervals. In particular, assume that the set of x satisfying the two constraints is a union of three intervals, the middle one of which we will call I . That is, the feasible set is like the shaded region in figure A2. Now what sort of conditions must hold at some point x^* in the interval I , if x^* is to be a candidate solution for the maximizing value of x ? (The fact that I is the middle of the three intervals is completely irrelevant to what follows.)

Consider first a case where x^* is in the interior of I , as in figure A3(a). Then, just as in the case of unconstrained maximization, it had better be that $f'(x^*) = 0$. Why? Because at such an x^* , we can move up a bit and

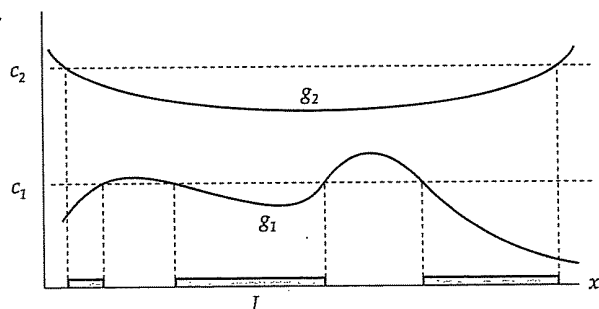


Figure A2. The feasible set.

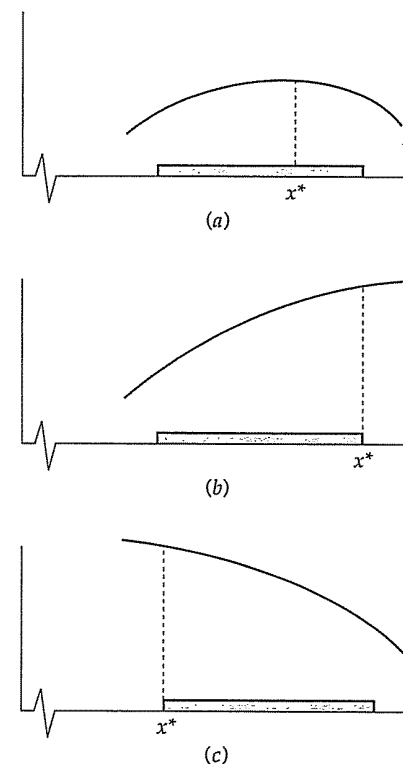


Figure A3. Possible locations of the optimal solution.

In (a), the solution, x^* is in the interior of a feasible interval. In (b), it is at a right-hand endpoint, and in (c) it is at a left-hand endpoint.

down a bit and not violate either constraint (that is, stay inside of I). This means that if x^* is going to be optimal, it has to be as good as values just above and just below it, and, as before, this implies that f cannot have a strictly positive or negative derivative at x^* .

Now suppose that x^* is up against the right-hand side of I , as in figure A3(b). Now x^* could be the optimal value of x even if $f'(x^*) > 0$;

we can increase the value of f by moving up a bit further, but this will violate one of the constraints.

And in figure A3(c) we see the other “boundary” possibility. If x^* is up against the left-hand side of the interval I , then $f'(x^*)$ could be negative and still x^* is a candidate for the optimum.

The FOCs, the CSCs, and the preceding pictures

Now we relate these three pictures to the FOCs and the CSCs. Let λ_i be the multiplier on the constraint $g_i(x) \leq c_i$ for $i = 1, 2$. Then the Lagrangian is $f(x) - \lambda_1 g_1(x) - \lambda_2 g_2(x)$, and the FOC is

$$f'(x) - \lambda_1 g'_1(x) - \lambda_2 g'_2(x) = 0.$$

Case a. Suppose we find a solution x^* to the FOC, the CSCs (and the constraints, including the nonnegativity constraints on the multipliers), such that neither constraint binds at x^* . That is, x^* is in the interior of an interval such as I , as in figure A3(a). Since x^* is in the interior of I , the CSCs mandate that $\lambda_1 = \lambda_2 = 0$, and the FOC will read that $f'(x^*) = 0$. Precisely what we said we’d want in such a case.

Case b. Suppose that x^* is a solution to all the equations and constraints, the first constraint binds at x^* (and the second is slack; recall our assumptions), and $g'_1(x^*) > 0$. Then x^* must be on the right-hand side of an interval such as I , as in figure A3(b). Why? Because since $g'_1(x^*) > 0$, increasing a bit past x^* will cause the constraint $g_1(x) \leq c_1$ to be violated, while decreasing a bit will cause this constraint to go slack (and since the other constraint is slack by assumption, it won’t be a problem, for small decreases in x , at least). Now the CSC allows λ_1 to be nonnegative; $\lambda_2 = 0$ must hold, however. Hence the FOC is now

$$f'(x^*) - \lambda_1 g'_1(x^*) = 0.$$

Since λ_1 is constrained to be nonnegative and $g'_1(x^*) > 0$ by assumption, the FOC reads that $f'(x^*) \geq 0$, which is precisely what we want for a candidate solution at the right-hand side of an interval; cf. figure A3(b).

Case c. Suppose the first constraint binds at x^* , the second is slack, and $g'_1(x^*) < 0$. Then by logic just as above, x^* will have to be on the left-hand side of an interval like I , complementary slackness will say that the FOC simplifies to $f'(x^*) - \lambda_1 g'_1(x^*) = 0$, and since $\lambda_1 \geq 0$ and $g'_1(x^*) < 0$, we have that $f'(x^*) \leq 0$, or just what we want for figure A3(c).

A few words about the coincidence in the example

About that coincidence, notice that in case b, in the solution to the FOC we’ll find that $\lambda_1 = f'(x^*)/g'_1(x^*)$. Now suppose we increase the right-hand side of the constraint $g_1(x) \leq c_1$ by just a bit — say, to $c_1 + \epsilon$. This is going to lengthen the interval I by a bit; the interval gets longer by approximately $\epsilon/g'_1(x^*)$. A picture may help you see this. In figure A4, we have a more detailed look at the situation in case b. On the bottom set of axes we’ve plotted g_1 , and you’ll see that the right-hand side of the interval I is determined by the solution of $g_1(x) = c_1$. When we increase c_1 by ϵ , we can push over the right-hand side, and to a first-order approximation, the right-hand side moves up by $\epsilon/g'_1(x^*)$. But then on the top set of axes, we see that loosening the constraint by this amount allows the function f to rise a bit more; it rises by approximately the amount $f'(x^*) \times (\epsilon/g'_1(x^*))$, which is $\lambda_1 \epsilon$. That is, the rate of increase in the objective function value at our “candidate solution” per unit increase in the right-hand side of

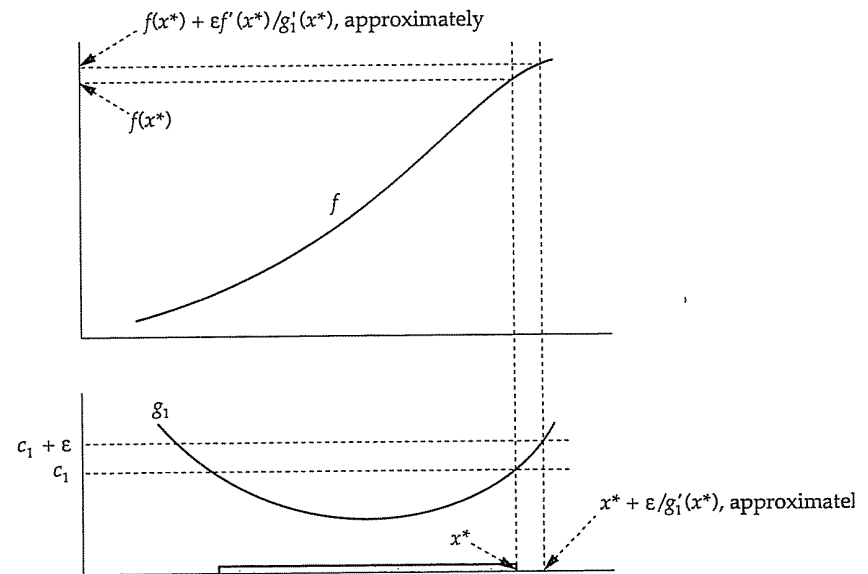


Figure A4. Explaining the coincidence.

the binding constraint is precisely the value of the multiplier on that constraint. This isn't special to case b; you could draw a similar picture for case c. (It might be a useful exercise to do so.) And this isn't anything special to the case of one dimension either, as long as the problem you're looking at is well behaved.

Some general comments

All this talk proves nothing. It is meant to indicate how the CSCs work — by modifying the FOCs in the “natural” way when you are at a point that is up against a constraint. The pictures are harder to draw when you move from one-dimensional to multidimensional problems, and I won't try to draw them here, but the same basic principles apply; complementary slackness says that a multiplier can be strictly positive only when a constraint binds, and then that multiplier is used to allow the objective function f to increase in directions that owing to the binding constraint(s) are infeasible.

Finally, you'll note that everything discussed in this section on intuition (after the review of unconstrained optimization) was about producing candidate solutions. Just as in unconstrained optimization, you might wonder what guarantee you have that a solution to all the equations and inequalities is really globally (or even locally) optimal. Accompanying theory about things analogous to the second-order conditions in unconstrained optimization gives you sufficient conditions for a candidate solution to be locally optimal. From this I happily spare you. And a theory analogous to the result in unconstrained optimization about concave f ensures that any candidate solution is in fact globally optimal. That theory is actually fairly simple, but I'll spare you that as well. However, I will assure you that for purposes of the problems in this book, at least until we get to oligopoly, any candidate solution is in fact a global optimum. And I'll even tell you how to watch for the exceptions in general: Watch for problems where the objective function f is not concave and/or the set of solutions of the constraints is not convex.

A1.4. Bibliographic notes

The cookbook recipe and very brief intuitive remarks presented here are certainly no substitution for learning properly about constrained optimization. Because constrained optimization plays such an important role in economic theory, the serious student of economics should study (either independently or in a course on mathematical methods) this subject. At a minimum, you should know about the FOCs and CSCs as necessary

conditions for an optimum and so-called *constraint qualifications*, *second-order conditions*, sufficiency of the FOCs and CSCs in *convex problems*, and problems with quasi-convex or -concave objective functions. Any number of books provide treatments of this material; Mangasarian (1969) covers much of the nuts and bolts, although this book doesn't explain why the multipliers are partial derivatives of the value function. Luenberger (1984) does discuss this and provides much of what one needs for convex problems but doesn't discuss quasi-convexity. Luenberger also provides an excellent treatment of linear programming and numerical methods for solving constrained and unconstrained programming problems.

References

- Luenberger, D. 1984. *Linear and Nonlinear Programming*, 2d ed. Reading, Mass.: Addison-Wesley.
- Mangasarian, O. 1969. *Nonlinear Programming*. New York: McGraw-Hill.

A1.5. Problems

- 1. What is the solution to the dieting consumer's problem if his calorie constraint is 1,650 calories? If his calorie constraint is 550 calories? If his calorie constraint is 1,600 calories?
- 2. Suppose that our consumer also has a problem with his cholesterol levels. Each unit of wheat supplies 10 unit of cholesterol, whereas each unit of candy supplies 100. The consumer has been told by his doctor to limit his cholesterol intake to no more than 260 units per day. What is the optimal solution to the consumer's problem now, with this additional constraint (with a calorie constraint of 1,550 calories per day)? Be sure to compute the value of the multipliers. Then work through the solution if the consumer (a) is allowed 261 units of cholesterol per day, 1,550 calories per day, and has \$10 to spend; (b) is allowed 260 units of cholesterol per day, 1,551 calories per day, and has \$10 to spend; and (c) is allowed 260 units of cholesterol per day, 1,550 calories per day, and has \$10.10 to spend. In terms of the coincidence we observed before, what happens in this case?
- 3. I've assured you that for most of the problems you encounter in this book, when you have one candidate solution, you will know that you have a global optimum. A part of this assurance may be provided by the following fact: If, in the general problem with which this appendix began, the function f is strictly concave, and the functions g_i are all convex, then the problem necessarily has at most a single solution. Prove this.

appendix two

Dynamic programming

For the solving of all manner of dynamic choice problems, an important technique is *dynamic programming*. In this appendix you will be briefly introduced to the techniques and basic results. All readers should go through the material in the first two sections, which concerns finite horizon problems. The material in the rest of this appendix is only used in optional (small type) parts of the book (mostly in chapter 14), and so can be regarded as optional as well. But the techniques of infinite horizon dynamic programming are becoming more pervasive in economics generally, and many of you will find that you have to learn this stuff some day. In case you find the treatment here too cryptic (which is quite possible, given the small amount of space devoted to the subject here), references to more expansive treatments are provided at the end of this appendix.

A2.1. An example with a finite horizon

We begin with an example of finite horizon dynamic programming. (This is problem 3[a] from chapter 4.)

A consumer lives for three periods denoted by $t = 0, 1, 2$ and consumes artichoke and broccoli each day. We let a_t be the amount of artichoke consumed on day t , and we let b_t be the amount of broccoli. This consumer has preferences over lotteries on consumption bundles (where a consumption bundle is a six-tuple $(a_0, b_0, a_1, b_1, a_2, b_2)$) that satisfy the von Neumann-Morgenstern axioms and that are represented by the von Neumann-Morgenstern utility function

$$U(a_0, b_0, a_1, b_1, a_2, b_2) = (a_0 b_0)^{.25} + .9(a_1 b_1)^{.25} + .8(a_2 b_2)^{.25}.$$

This consumer can buy artichoke and broccoli in the marketplace each day. Because both vegetables spoil rapidly, what he buys on any given day is his consumption for that day. (He is constrained to consume nonnegative

amounts of each vegetable.) The price of artichoke is \$1 per unit on each and every day. The price of broccoli is more complex: It begins as \$1 per unit of broccoli at $t = 0$. But at $t = 1$, it is either \$1.10 or it is \$.90, each of these being equally likely. And at $t = 2$, the price of broccoli is again random and depends on the price the day before: If the price was \$1.10 at $t = 1$, then it is either \$1.20 or \$1.00, with each equally likely. If the price was \$.90 at $t = 1$, then at $t = 2$ it is either \$.98 or \$.80, each equally likely. At date $t = 0$, the consumer has no information (beyond what is given above) about subsequent prices of broccoli. At date $t = 1$, he knows the current price of broccoli and no more. At date $t = 2$, he knows the current price.

This consumer has \$300 to spend on artichoke and broccoli over the three days, which he can divide any way he wishes. Any money he doesn't spend on a given day sits in his pocket where it earns zero interest.

This is a fairly complex problem, because the consumer has a large number of decisions to make. He must decide how much artichoke and broccoli to consume at $t = 0$ (and, implicitly, how much of his \$300 to save); given the price of broccoli at $t = 1$ he must decide how much of what remains of his \$300 to save and how much to spend immediately on artichoke and broccoli, and he must decide how much to spend on artichoke and broccoli at $t = 2$, given prices at $t = 2$. Even if we assume that any money unspent at $t = 0$ and $t = 1$ is saved (which is certainly a reasonable assumption) and that all remaining money will be spent on broccoli or artichoke at $t = 2$ (also reasonable), this gives us two variables for $t = 0$, four variables for $t = 1$ (two vegetable amounts in each of two contingencies), and four variables for $t = 2$ (one vegetable amount in each of four contingencies). Rather than try to optimize over all ten variables at once, we will approach the problem in steps.

Step 1. Find contingent optimal decisions for $t = 2$. To begin, imagine that the consumer arrives at time $t = 2$ with z remaining to be spent. This consumer must decide how to allocate that z between artichoke and broccoli. Because his utility function U is additively separable over time periods, this decision can be made without reference to the earlier consumption decisions the consumer made. The consumer will solve the problem

$$\begin{aligned} &\text{Given } z, \text{ maximize } .8(a_2 b_2)^{.25} \\ &\text{subject to } a_2 + p_b b_2 \leq z, \end{aligned}$$

where p_b is the current price of broccoli. This is a simple maximization problem, and if you go through the algebra, you'll find the solution

$$a_2 = \frac{z}{2} \quad \text{and} \quad b_2 = \frac{z}{2p_b}.$$

That is, the expenditures on the two types of vegetables are the same.¹

From this we are able to compute the value of the objective function as a function of z for each of the four possible prices of broccoli. Substituting the optimal values of a_2 and b_2 , the optimal value of $.8(a_2 b_2)^{.25}$ is

$$.8 \left[\frac{z}{2} \right]^{.5} \left[\frac{1}{p_b} \right]^{.25},$$

which is (approximately) $.54048z^{.5}$ if $p_b = 1.2$, $.56569z^{.5}$ if $p_b = 1$, $.56855z^{.5}$ if $p_b = .98$, and $.59814z^{.5}$ if $p_b = .8$.²

We have determined the optimal *contingent* decisions at time $t = 2$, so-called because these decisions are contingent on two things: The price that prevails for broccoli, and the amount of money left to buy vegetables at $t = 2$.

Step 2. Find the optimal contingent decisions for $t = 1$. Now we step back to decisions that are made at $t = 1$. Imagine that the consumer has y left to spend at this point. He must divide this between immediate consumption of artichoke and broccoli and money left for $t = 2$. From step 1, we know how much, on average, money left for $t = 2$ will be worth, assuming the consumer purchases vegetables optimally at date $t = 2$. Hence we can use that value in determining what decisions to make at $t = 1$.

Let us illustrate: Suppose the price of broccoli is \$.90, so that next period the price will be either \$.98 or \$.80, each equally likely. The consumer must divide his y between artichoke, broccoli, and money left for $t = 2$. If we let a_1 , b_1 , and z be these three amounts, the expected utility

¹ If you have never worked with this sort of utility function before, and this solution seems remarkable to you, you should definitely do the algebra leading to this conclusion. This sort of utility function, which is part of the Cobb-Douglas family of utility functions, is much used in examples in economics. In fact, you might find it helpful to solve the problem $\max a^\alpha b^\beta c^\gamma$ subject to $p_a a + p_b b + p_c c \leq y$. Also try maximizing $\alpha \ln(a) + \beta \ln(b) + \gamma \ln(c)$ subject to the same constraint.

² These are, of course, indirect utility functions, and you should think of them as such if it helps your understanding.

for the consumer of the decision (a_1, b_1, z) (for periods $t = 1$ and $t = 2$ together) is

$$.9(a_1 b_1)^{.25} + [.5(.56855z^{.5}) + .5(.59814z^{.5})].$$

This is the crucial step, so let us take this slowly. The first term in the expression is the utility derived from the amounts of vegetable consumed immediately; and the second term (everything within the square brackets) is the expected utility derived from saving z until next period. The two .5s are the probabilities of the two different prices next time, and the terms inside the parentheses are the utilities derived from z in each of the two contingencies. So at $t = 1$, if the price of broccoli is \$.90, the consumer will pick (a_1, b_1, z) to maximize the function just displayed, subject to the constraint that

$$a_1 + .9b_1 + z \leq y.$$

Similarly, if the price of broccoli is \$1.10, so the possible prices of broccoli next period are \$1.20 and \$1.00, the consumer will (at $t = 1$, if he has y with which to consume) solve

$$\max .9(a_1 b_1)^{.25} + [.5(.54048z^{.5}) + .5(.56569z^{.5})] \text{ subject to } a_1 + 1.1b_1 + z \leq y.$$

You are certainly capable of solving these two problems, although it may take a fair bit of scribbling on some scratch paper. We will give a trick for solving these two problems in the next paragraph (in small type), but this trick (while it employs dynamic programming in a perverse way) will probably confuse you more than help if you are seeing dynamic programming for the first time. So let us be clear that the key step to dynamic programming is what you have just seen and not what will follow in small type. The key step is that we solved the period $t = 2$ problem as a function of z and the price of broccoli, and then took the value of the optimal solution (as a function of z and the possible prices of broccoli) and employed it in determining what to do at date $t = 1$. Because we have already found the optimal way to spend z at $t = 2$, we don't need to include that explicitly in our $t = 1$ problems; we only need the optimal values as a function of z . We proceed to solve the two optimization problems posed above, by hook or by crook, which then gives us optimal value functions for consuming from $t = 1$ on to the end as a function of y and the price of broccoli at $t = 1$; after the small type we will use these

A2.1. An example with a finite horizon

value functions to solve the $t = 0$ problem facing the consumer. For the record, if you solve the two optimization problems given above, you get the following answers:

If the price of broccoli is \$.90, set $a_1 = .27822y$, $b_1 = .30914y$, and $z = .44355y$, which gives expected utility equal to $.87590y^{1/2}$.

If the price of broccoli is \$1.10, set $a_1 = .27899y$, $b_1 = .25363y$, and $z = .44201y$, which gives expected utility equal to $.83190y^{1/2}$.

How did we get these answers? The trick we employed was to look at the $t = 1$ problem in two pieces. Fix a division of the total resources y between immediate consumption and future consumption, say $y - z$ for now and z for later. Optimally divide the $y - z$ between immediate broccoli and artichoke. This is a simple problem; it shouldn't take you long to see (if you don't already know) that half of $y - z$ will be spent on each of the two vegetables, leading to $a_1 = (y - z)/2$ and $b_1 = (y - z)/(2p_b)$. This means that $y - z$ allocated for current consumption yields immediate utility

$$.9[(y - z)^2/(4p_b)]^{.25} = .9[1/p_b]^{.25}[(y - z)/2]^{.5}.$$

All this is for a fixed division of y into immediate and future consumption: We optimize the split by maximizing

$$.9[1/p_b]^{.25}[(y - z)/2]^{.5} + k(p_b)z^{.5},$$

where $k(p_b)$ is a constant which we take from whichever of the two objective functions we are working with; .58334 for $p_b = .90$ and .55308 for $p_b = 1.10$. If you do the algebra, you'll find these to be particularly simple optimization problems, leading to the answers we gave above.

Step 3. Find the optimal decisions for $t = 0$. Finally we get to $t = 0$. If we let a_0 be the amount of artichoke the consumer buys immediately, b_0 the amount of broccoli, and y the amount he saves, and if we assume that, contingent on the price of broccoli at $t = 1$, the consumer proceeds optimally with the y that is in his pocket, he will have expected utility

$$(a_0 b_0)^{.25} + [.5(.87590y^{.5}) + .5(.83190y^{.5})].$$

Once again, this is the sum of the immediate utility of consumption and the expected value provided by having y in hand beginning next period,

where we average over the two possible prices. This objective function must be maximized subject to the constraint

$$a_0 + b_0 + y \leq 300,$$

and if you carry out the maximization, you will find that the solution is

$$a_0 = b_0 = 61.02 \text{ and } y = 177.96.$$

Now that we know the optimal value of y , we can reconstruct the optimal levels of consumption at time 1, depending on the price of broccoli. For example, if the price of broccoli at $t = 1$ is \$1.10, set

$$a_1 = (.27899)(177.96) = 49.65, b_1 = (.25362)(177.96) = 45.14, \text{ and } z = (.44202)(177.96) = 78.66.$$

And with this, you can work out how to consume at date $t = 2$, depending on the price of broccoli.

A2.2. Finite horizon dynamic programming

We hope that the intuition behind this solution technique is fairly clear. We compute the optimal thing to do at the last date, contingent on conditions that apply. This gives us the expected value of carrying forward any amount of money from date $t = 1$ and then proceeding optimally, which we use to compute the optimal thing to do at $t = 1$, contingent on conditions then applying. And this gives us the expected value of carrying forward any amount of money from date $t = 0$ and then proceeding optimally, which we use to compute the optimal thing to do at time $t = 0$.

This technique works in all manner of problems besides consumption-savings problems. Here is another typical (and somewhat simpler) example. You are searching for employment. You have four interviews lined up today, and for reasons we will not go into, you will definitely take one of the four jobs. You know (given your sterling credentials) that each firm with which you interview will offer you a job; what is unclear is the wage they will offer. We let w_t be the wage offered by firm t , for $t = 1, 2, 3, 4$. You interview at each firm in turn, and at the end of the interview the firm offers you a wage. *If you refuse the job when it is offered, you cannot go back to it.* You must decide on the spot whether to take a particular job. Wage offers are all uniformly distributed on the interval $[0, 10]$ and are

statistically independent of each other. You wish to maximize the expected wage you will be paid.

This is an example of an optimal stopping problem, so called because the sole decision you make is whether to "stop" the process or continue at each stage. It is used in modeling job search, searching for the lowest price at which to buy a good, searching for the highest quality individual among those being interviewed, deciding when to harvest a crop, and so on. The assumption that you are unable to recall a previous offer is, perhaps, a bit unnatural in the context we've described, and you might think of a somewhat more complex model where, if you refuse a job offer, you can later go back to it, but then there is a chance it will have been filled by someone else. (See problem 3.) Or you can think of a situation in which you must pay a cost every time you interview at another firm, but you can always go back to the best offer made so far. (See problems 1 and 2.)

We solve this problem using the same sort of technique we used in the previous section, by going to the last period and then working through to the first. If you haven't taken a job by the time you will interview at firm 4, you will be forced to take their offer. This gives you an expected wage of 5.

Hence when you hear the offer at firm 3, your decision is simple. If you are offered a wage below 5, you do better to continue on to firm 4, which gives you an expected wage of 5. But if you are offered a wage above 5, you do better to take this.

What, then, is your expected wage from following an optimal strategy at firms 3 and 4, given you didn't take a job at either firm 1 or firm 2? It is

$$\int_0^{10} \max\{w, 5\}(.1)dw.$$

This is obtained by averaging over all the possible values of w_3 the expected wages you will receive by acting optimally: If $w_3 < 5$, then your expected wage is 5 (because you continue on to firm 4), while if $w_3 \geq 5$, your expected wage is w_3 .³ If we evaluate this integral, we get

$$\begin{aligned} \int_0^{10} \max\{w, 5\}(.1)dw &= \int_0^5 \max\{w, 5\}(.1)dw + \int_5^{10} \max\{w, 5\}(.1)dw \\ &= \int_0^5 5(.1)dw + \int_5^{10} w(.1)dw = \frac{1}{2}5 + \frac{.1w^2}{2} \Big|_5^{10} = 2.5 + 3.75 = 6.25. \end{aligned}$$

³ The density of a uniform distribution on $[0, 10]$ is .1 for all $w \in [0, 10]$.

That is, if you turn down the jobs offered at firms 1 and 2 and proceed optimally, your expected wage will be 6.25.

So now it is clear what you should do upon hearing job offer 2. If this offer exceeds 6.25, take it. If it falls below 6.25, turn it down and take your chances with firms 3 and 4. And so proceeding optimally beginning with firm 2 gives you an expected wage of

$$\begin{aligned}\int_0^{10} \max\{w, 6.25\}(.1)dw &= \int_0^{6.25} \max\{w, 6.25\}(.1)dw + \int_{6.25}^{10} \max\{w, 6.25\}(.1)dw \\ &= \int_0^{6.25} 6.25(.1)dw + \int_{6.25}^{10} w(.1)dw \\ &= \frac{5}{8}6.25 + \frac{.1w^2}{2} \Big|_{6.25}^{10} = 3.90625 + 3.046875 = 6.953125.\end{aligned}$$

And now it is clear what you should do upon hearing job offer 1. If this offer exceeds 6.953125, take it. If it is less, then turn it down and take your chances with jobs 2, 3, and 4. (What is the expected wage you garner by following the optimal strategy in this case? If you work through the math, you should find that it is approximately 7.4173.)

Once again, the idea is that we solve the problem from the back. We work out what the optimal strategy and optimal value are starting from each time until the end recursively, where the optimal value starting at $t+1$ is used to find the optimal strategy and optimal value starting at time t . We are using the basic principle that

If a strategy is optimal for each point in time at that point of time, given that an optimal strategy will be used thereafter, then the strategy is optimal.

This is known as the *principle of optimality*, or as *Bellman's principle of optimality*, or as the *discrete maximum principle*.

Underpinning this principle is the following simple bit of mathematics. Suppose we wish to maximize a function of two variables, $f(x, y)$. For each x , we can find that value of y that maximizes $f(x, y)$ holding x fixed; and then we can find the overall maximum by optimizing over x . That is, first compute $f^*(x) = \max_y f(x, y)$, then maximize f^* over x . In symbols,

$$\max_{x,y} f(x, y) = \max_x [f^*(x)] = \max_x [\max_y f(x, y)]. \quad (*)$$

If we were being very precise about this, we'd worry about whether those maxima are achieved^a and we'd worry, as in our first application, about how to handle constraints on the variables x and y . But the basic idea is conveyed by (*).

There is nothing in (*) about the decision concerning y being made "after" the decision concerning x ; equation (*) is phrased for a general two-variable optimization problem. So it is with dynamic programming. Most applications, and certainly most applications in economics, involve decisions made through time. In such cases, it is natural to work from later dates back to earlier dates. This is especially true in applications with uncertainty that resolves through time, because then later decisions are conditioned on more information. But from the point of view of the optimization theory being employed in dynamic programming, this isn't necessary. If, for example, there was no uncertainty in our artichoke and broccoli problem, then there would be nothing to prevent us from first finding the optimal way to split z between artichoke and broccoli at $t=1$, then computing the optimal way to split y between artichoke and broccoli at $t=0$ and funds for consumption at time $t=1$, and then computing the optimal way to split the entire \$300 between the two types of vegetable at $t=2$ and funds for consumption at $t=0$ and $t=1$. It might not be natural to think of the problem this way, but the optimization would go through just the same.

We will, for the remainder of this appendix, speak in terms of decisions made through time and the use of dynamic programming to solve such optimization problems. But you should not be misled into thinking that to employ dynamic programming it is necessary that decisions have this sort of temporal ordering.

Things that link the past, the present, and the future

A difference between the consumption-savings problem of the previous section and the job-taking problem described in this section is that in the consumption-savings problem, when we analyze decisions made at later dates, we analyze those decisions depending on the level of money in the consumer's pocket (z or y) and the current price of broccoli. The amount of money in the consumer's pocket is a variable that links past decisions with present opportunities, and it links the present decision with future opportunities, and thus future values. The price of broccoli at any point in time determines (in part) current opportunities. But it also is used to "predict" future broccoli prices, and so it provides a further link between the past and present on one hand and the future on the other.

In general dynamic programming problems, the opportunities the decision maker has now and will have in the future depend on previous decisions made. And when there is uncertainty about the future, data

^a Is (*) correct if we replace the *maxs* with *sup*s?

from the past and present are useful in predicting the future. These links must be included in the analysis; optimal decisions and values must be computed contingent on whatever links are pertinent.

In this regard, note two things about the formulation of the consumption-savings problem used in the previous section.

(1) In general, one of the things that can link past with present and future is that past decisions can affect the preferences one has over current decisions. It is not hard to imagine an individual whose preferences mandate that excessive artichoke at $t = 0$ lessens the utility derived from artichoke at $t = 1$ and $t = 2$. The preferences that we specified did not have this property; because utility is additively separable through time, the marginal rates of substitution of broccoli for artichoke at any time beyond a given date are independent of consumption up to that date.

What would happen if we had preferences where the levels of consumption at one date affected the marginal rates of substitution at later dates? Then in solving for the optimal way to spend z at date $t = 2$, we would have to condition on past consumption levels. That is, we would compute "values of optimally continuing" as a function of z and the price of broccoli, as before, and also as a function of the vector (a_0, b_0, a_1, b_1) . Depending on the exact specification of the consumer's preferences, we might be able to get away with something less than conditioning on the entire vector of past consumption. For example, if all that matters (in terms of previous consumption) to preferences for consumption at date t and beyond is the sum total consumption of artichoke, then we could condition our date 2 decisions on $a_0 + a_1$ and our date 1 decision on a_0 only.

In most applications in economics, preferences are assumed to be additively separable in just the fashion we've assumed here. (Additive separability is a bit more than is needed, but it is convenient parametrically.) The point to understand here is that this is a matter of analytical convenience and not necessity. In theory, one can use dynamic programming without additive separability as long as all the "links" between past, present, and future are properly taken into account. (For most applications, though, additive separability is a practical necessity, because otherwise the analysis becomes intractable.)

(2) Imagine that we reformulated the consumption-savings problem so that it has four time periods, and so that at $t = 0$, the price of broccoli is \$1; at $t = 1$, it is either \$1.10 or \$.90; at $t = 2$, it is either \$1.20, \$1.00, or \$.80; and at $t = 3$, it is \$1.30, \$1.10, \$.90, or \$.70. Imagine sitting at time $t = 2$ with a broccoli price of \$1.20 and y in the consumer's pocket to spend

immediately or to save for period 3. The expected utility of funds saved depends on the future price of broccoli, and so the decision taken at $t = 2$ depends on the best current assessment of that future price.

It is easy to imagine a formulation of the evolution of the price of broccoli where the only relevant data for predicting future prices of broccoli is the current price of broccoli. For example, we can imagine a model where the price of broccoli rises or falls by a \$.10 in each period, with each of those two changes equally likely. In this case, the decision made at time $t = 2$ would depend on the current price of broccoli in two ways: This price determines what bundles of current vegetable are affordable, and it determines the distribution of the value of money set aside for future consumption.

But it is also possible to imagine formulations where best assessment that can be made at time $t = 2$ about the price of broccoli at time $t = 3$ depends on both the current price of broccoli and the price in the previous period. (Suppose broccoli takes two periods to harvest, and farmers who planted at time $t = 1$ determined how much to plant in part by looking at the prevailing price of broccoli.) In this case, in deciding how to split y at time $t = 2$ between current and future consumption, we have to condition the decision on the previous price of broccoli, as well as the current price, insofar as the previous price helps us to assess future prices.

If we had such a formulation, there would be no problem theoretically in using dynamic programming. Decisions made at $t = 2$ would be conditioned on previous prices of broccoli, as well as on current prices and the amount of money the consumer possesses.

As a practical matter, however, this increases the complexity of the problem by increasing the "number" of contingent problems that must be solved for $t = 2$. The analytical practicalities lead economists to formulate models in which the past is useful for predicting the future in particular low-dimensional ways. That is (if you know the terminology), models usually assume that there are low-dimensional statistics of the past that are sufficient for predicting the future, and these low-dimensional sufficient statistics become "state variables" or links in the dynamic programming formulations. Indeed, in many formulations it is assumed that current prices are sufficient for predicting the future; the fancy way of saying this is that *Markovian* assumptions are made.

A2.3. An example with an infinite horizon

While two examples certainly don't constitute a general theory, the two examples given will probably give you enough lead on how to use

dynamic programming in finite horizon problems. A bit more complex is the use of dynamic programming techniques for infinite horizon problems. That is, we imagine a formulation in which there is no "last decision" from which we can begin our analysis; at each point in time there are always many more decisions to come. We look at a specific example in this section; the general theory is discussed in the next.

Consider the following variation on the consumption-savings problem of section A2.1. (This is problem 3[b] from chapter 4.) The consumer now consumes at times $t = 0, 1, 2, \dots$. At each date, he consumes both artichoke and broccoli. His utility function, defined for an infinite stream of consumption $(a_0, b_0, a_1, b_1, \dots)$ is

$$U(a_0, b_0, a_1, b_1, \dots) = \sum_{t=0}^{\infty} (.95)^t (a_t b_t)^{.25}.$$

This consumer starts with wealth \$1,000, which he can spend on vegetables or can deposit in his local bank. Any funds deposited at the bank earn interest at a rate of 2% per period, so that \$1 deposited at time t turns into \$1.02 at time $t + 1$.

The price of artichoke is fixed at \$1. The price of broccoli is random. It begins at \$1. Then, in each period, it either increases or decreases. Being very specific, if the price of broccoli at date t is p_t , then

$$p_{t+1} = \begin{cases} 1.1p_t & \text{with probability .5, and} \\ .9p_t & \text{with probability .5.} \end{cases}$$

The consumer wishes to manage his initial wealth \$1,000 in a way that maximizes his expected utility. How should he do this?

We will "solve" this problem by using the following recipe.

Step 1. Guess what strategy our consumer should follow in this problem. Call your guess the candidate strategy.

Step 2. For the candidate strategy, compute the expected discounted utility the consumer will obtain if he follows this strategy as a function of his current level of wealth and the current price of broccoli.

*Step 3. Having computed the value of following the candidate strategy, ask the question, Can the consumer do any better by a strategy that is completely arbitrary at the start and then conforms forever after to the candidate strategy? If the answer is no, the candidate strategy is said to be **unimprovable**, where we should really say that it is unimprovable in a single step. If the strategy is not*

unimprovable, then it is not optimal. And, for a wide range of infinite horizon problems (including this problem), unimprovable strategies are optimal.

This recipe is not, to be sure, very transparent, and you shouldn't despair that you don't understand it until after you've seen it worked out on an example (which happens momentarily). But it is probably sufficiently comprehensible so that you can see that it isn't really a recipe for solving the problem directly. If you have a guess what the solution is, this recipe allows you to check whether your candidate strategy is in fact optimal.

Now try this recipe out on our problem.

Step 1. Guess what strategy our consumer should follow in this problem. We guess that the strategy will take the following form: In each period the consumer spends a constant fraction of his wealth on current consumption, spending half that amount on artichoke and half on broccoli, with the rest of his wealth going into savings. We don't know what the fraction is, so for the time being we make it a variable, namely k .

Where did this guess come from? Since the solution to this problem is not original to this book, the accurate but not very useful answer is that it appears in the literature. But if it hadn't, some techniques lead to this guess; techniques that we will explore at the end of the next section. For now, assume that this guess is the result of brilliant inspiration.

Step 2. For the candidate strategy, compute the expected discounted utility the consumer will obtain if he follows this strategy as a function of his current level of wealth and the current price of broccoli. We perform this computation as follows: Let us write $v(y, p)$ for the value to the consumer of following the strategy just described if his initial wealth is y and the price of broccoli begins at p . (The initial conditions we are given are $y = 1,000$ and $p = 1$, but we take a slightly more general approach here.) And, to make the equations a bit prettier, let us write $\hat{k}$ for $1 - k$. Then $v(y, p)$ satisfies the functional equation

$$v(y, p) = \left[\frac{ky}{2} \frac{ky}{2p} \right]^{.25} + .95 [.5v(1.02\hat{k}y, 1.1p) + .5v(1.02\hat{k}y, .9p)].$$

This comes from the following considerations. If the guessed-at strategy is followed, and if we begin with initial conditions (y, p) , then the strategy calls for setting immediate consumption of artichoke equal to $ky/2$ and immediate consumption of broccoli equal to $ky/(2p)$. The immediate utility from this is just the first term on the right-hand side. This strategy also means that $\hat{k}y$ will go into the bank, and so $1.02\hat{k}y$ will come out

next period. The price of broccoli next period will be either $1.1p$ or $.9p$, each equally likely, and since the strategy we are following looks the same starting next period (as a function of the then initial conditions) as it does now, the value of following the strategy, discounted back to next time, is either $v(1.02\hat{k}y, 1.1p)$ or $v(1.02\hat{k}y, .9p)$, with each of these equally likely. The second term on the right-hand side is thus the expected value of the utility of future consumption (if we follow this strategy), where the extra .95 discounts this expected value back to the current period.

This is a pretty miserable looking functional equation, but now for the next bit of magic: If the guessed at strategy is followed, then $v(y, p)$ will have the form $Ky^{.5}/p^{.25}$ for some constant K . How do we know this? Think of it this way: If the consumer's initial wealth is doubled, then the strategy calls for him to eat precisely twice as much artichoke and broccoli this period, and his wealth next period will be precisely twice what it otherwise would have been, so in the next period he doubles his consumption, and so on. Given the consumer's utility function, this means that his utility goes up by the factor of $2^{.5}$. Similarly, if the price of broccoli were doubled, the consumer would spend the same amount of money on artichoke and broccoli in each period, but he would get half the broccoli he would have had otherwise, which, given his utility function, means his utility decreases by the multiplicative factor $1/2^{.25}$. (These are loose verbal arguments but they contain the germ of a formal proof; you are asked to provide a rigorous proof in problem 5.)

Once we know that this is the form of $v(y, p)$, we can substitute into the functional equation we had before, and we get the equation

$$K \frac{y^{.5}}{p^{.25}} = \left[\frac{ky}{2} \frac{ky}{2p} \right]^{.25} + .95 \left[.5K \frac{[1.02\hat{k}y]^{.5}}{[1.1p]^{.25}} + .5K \frac{[1.02\hat{k}y]^{.5}}{[.9p]^{.25}} \right].$$

This doesn't look too much prettier, but looks are deceiving: It has its virtues. For one thing, the term $y^{.5}/p^{.25}$ can be factored out of both sides, leaving an equation in the variables K and k . (Recall that $\hat{k} = 1 - k$, so $\hat{k}$ is not another variable.) Taking that equation, a bit of fairly simple algebra will allow you to isolate K on one side of the equation; we obtain (approximately)

$$K = \frac{(k/2)^{.5}}{1 - .96096132\hat{k}^{.5}}. \quad (\clubsuit)$$

Note what we've done: We considered a strategy of a particular form, parameterized by k , the fraction of wealth that is used for current consumption. Because of the simple nature of the strategy, we are able to

prove (or, more precisely, you will prove for homework) that the expected utility of using that strategy has a particular functional form, up to a second parameter K . We could also find a functional equation of the expected utility of using the strategy, and from this we are able to find what value K takes as a function of parameter k .

It may be worth noting that this function, given by $(\clubsuit)$ behaves rather well at the two extreme values of k , namely $k = 0$ and $k = 1$. If $k = 0$, then the consumer always invests all of his wealth and never has anything to eat. Of course, this leads to $K = 0$; if you never eat, you never get any utility. And if $k = 1$, then $\hat{k} = 0$, and $K = (1/2)^{.5}$; our consumer spends his entire endowment on initial consumption and, of course, this gives him utility $(y/2)^{.5}(1/p)^{.25}$.

One thing we can do immediately is to answer the question, Of all the strategies of the form of our guess, which value of k gives the highest lifetime utility? That is, which value of k gives the highest value of K ? Taking the derivative of the function giving K in terms of k doesn't look particularly appealing, so we approach this problem numerically. Up to six decimal points, the K is maximized at $k^* = .076553$, for a value of $K^* = 2.55565129$. Note well: This doesn't show that the candidate strategy for k^* is the optimal strategy overall. All we've shown so far is that among strategies of this special type, k^* is best. (Indeed, it is only now that we have truly completed step 1. Until now we guessed at the form of the optimal solution. Now at last we have a single candidate strategy.)

But now we show that the strategy with k^* is optimal.

Step 3. Having computed the value of following the candidate strategy, ask the question, Can the consumer do any better by a strategy that is completely arbitrary at the start and then conforms forever after to the candidate strategy? If the answer is no, the candidate strategy is said to be unimprovable, where we should really say that it is unimprovable in a single step. If the candidate strategy is not unimprovable, it is not optimal. And for a wide range of infinite horizon problems (including this problem), unimprovable strategies are optimal.

The last line is the punchline, so please read it again. If the candidate strategy passes the test that it cannot be improved upon in a single step, then it is optimal. This isn't true for every problem in dynamic programming; but sufficient conditions will be supplied below, and for the time being you can rest content in the knowledge that this problem meets those conditions. If our strategy passes the unimprovability test, then we've got a winner.

Suppose, then, that the consumer begins with wealth y , when the price of broccoli is p . The immediate decisions to make are how much

to spend on artichoke, how much on broccoli, and how much to save. Let a be the amount of artichoke consumed immediately, b the amount of broccoli, and z the amount saved. Then the utility the consumer obtains from these three immediate decisions followed by conformance to our candidate solution is

$$(ab)^{.25} + .95 \left[.5K^* \frac{(1.02z)^{.5}}{(1.1p)^{.25}} + .5K^* \frac{(1.02z)^{.5}}{(.9p)^{.25}} \right].$$

This is the sum of the utility of immediate consumption and the expected utility from future consumption, discounted back to the current period. Note carefully that the candidate solution is buried in second term through the multiplicative factor K^* . More generally, what we have inside the brackets in the second term is the expected value of the value of the candidate solution, which we computed in step 2, averaged over the possible initial conditions beginning next period, given current conditions and current decisions. We wish to maximize this expression subject to the budget constraint $a + pb + z \leq y$, and we hope that the solution to this optimization problem is

$$a = k^*y/2, b = k^*y/(2p), \text{ and } z = (1 - k^*)y.$$

And, sure enough, if you plug in $K^* = 2.55565912$, the optimal solution has $a = .076553y/2$, and so on. Up to the limits of our computational ability, the candidate strategy is unimprovable in a single step. (And if we ground through the problem algebraically, we would find that the precise candidate solution is unimprovable.) Hence, by the test of step 3, our candidate solution is optimal.

A2.4. Stationary Markov decision problems

This is all very nice, but you may wonder why this test works, and how we might pose all this a bit more generally. (Also left in limbo is the question of how we ever managed to guess at the nature of the solution.) We will not give proofs, but by developing the theory just applied in greater generality, we can at least give the basic intuition.

Formulation of the general problem

We imagine a decision maker who must take an action α_t at each of a countable number of dates, $t = 0, 1, 2, \dots$. A current "state of affairs"

prevails at each time t , denoted by the variable θ_t .^b The state θ_t serves three functions. First, it determines which actions are feasible at the current date; we write $A(\theta_t)$ as the set of actions feasible at time t as a function of the state θ_t . Second, together with the chosen action, it determines a reward that the decision maker receives at time t ; we write $r(\theta_t, \alpha_t)$ for this reward. Third, it serves as a sufficient statistic of all history prior to date t for predicting the values of future state variables. If the decision maker takes action α_t at time t , the probability that the next state is θ_{t+1} (given all information available up to time t) is $\pi(\theta_{t+1}|\theta_t, \alpha_t)$.

The decision maker's problem is to select actions at each time as a function of past history to maximize the expected discounted sum of rewards he receives; that is, he seeks to maximize the expectation of

$$\sum_{t=0}^{\infty} \gamma^t r(\theta_t, \alpha_t),$$

where future states (and, therefore, future actions) are random. We assume that $\gamma \in [0, 1)$.

This sort of problem is called a *stationary Markov decision problem*. The modifier "Markov" is used because the current state is a sufficient statistic for predicting future states (given the current action). The modifier "stationary" is used because the problem facing the decision maker looks the same (contingent on the state) at each time t .

Our consumption-savings problem fits into this general framework (with one bit of shoehorning). Think of $\alpha_t = (a_t, b_t, z_t)$, where a_t is the amount of artichoke consumed at time t , b_t is the amount of broccoli consumed, and z_t is the amount of money saved for next period. For the state variable, think of $\theta_t = (y_t, p_t)$, where y_t is the amount of money available at the start of period t and p_t is the current price of broccoli. Then:

- (1) $A(\theta_t) = \{(a_t, b_t, z_t) : a_t + p_t b_t + z_t \leq y_t, a_t, b_t, y_t \geq 0\}$; these are just the usual budget and nonnegativity constraints;
- (2) $r(\theta_t, \alpha_t) = (a_t b_t)^{.25}$ for $\alpha_t = (a_t, b_t, z_t)$; that is, the immediate reward in any period is the utility of consumption in that period; and

^b For reasons of analytical tractability, we assume that there are at most countably many possible values of θ_t . We won't explain why this assumption is made, except to say that we don't want the mathematical results we will announce in a bit to be incorrect, and without this technical condition one must worry about the measurability of some of the functions we define. It is possible to do without this assumption and, as we will note later, we must do without it to accommodate the example of the previous section, but then we would have to make a larger number of technical assumptions in footnotes similar to this one.

(3) $\theta_{t+1} = (y_{t+1}, p_{t+1})$ is determined by $\theta_t = (y_t, p_t)$ and $\alpha_t = (a_t, b_t, z_t)$ as follows: $y_{t+1} = 1.02z_t$ with certainty, and p_{t+1} has probability distribution (conditional on p_t) given by

$$p_{t+1} = \begin{cases} 1.1p_t & \text{with probability .5, and} \\ .9p_t & \text{with probability .5.} \end{cases}$$

Finally, $\gamma = .95$.^c

Strategies and value functions

A *strategy* for the Markov decision problem is a rule for choosing actions at each time t as a function of all the information available at time t , namely all previously taken actions and the present and all previous states. We write σ as a symbolic representation of a strategy, which is really quite a complex animal. Each strategy for choosing actions together with an initial state θ_0 leads to a probability distribution over the sequence of rewards the decision maker will receive, and so for each strategy σ we can (in theory) compute a function that gives the expected discounted sum of rewards the decision maker will receive by following σ as a function of the initial state θ_0 . We denote this expected value by $v(\theta_0, \sigma)$.^d

In general, it will be very hard to compute the expected value of following a strategy. But for some strategies the computation is not so difficult. The strategy σ is said to be *stationary* if the action prescribed by σ at time t when the current state is θ_t depends *only* on θ_t . The "only" here has two parts to it: The action chosen doesn't depend on the current date *per se*, and it doesn't depend on anything that happened prior to time t .

For a stationary strategy σ , we can build a *recursive equation* for the value function associated with using σ . We will write this equation and then explain it. One piece of notation is needed: When σ is a stationary strategy, we write $\sigma(\theta)$ for the action prescribed by σ when the state is θ . Then the recursive equation is

$$v(\theta_0, \sigma) = r(\theta_0, \sigma(\theta_0)) + \gamma \sum_{\theta} v(\theta, \sigma) \pi(\theta | \theta_0, \sigma(\theta_0)).$$

^c There is one problem with this. The range of possible values of θ_t is uncountable, because there are uncountably many savings decisions that can be made. For this reason, the results we announce don't quite apply to our problem. Or, to be more precise, the results are correct, but it takes a bit more on the technical side to be sure that this is so.

^d Some technical restrictions are needed to be sure the integrals that define this expectation are well defined. If r is bounded above or below, then the integrals are well defined, although values of minus or plus infinity, respectively, are not precluded.

Note that this is a recursive equation because the unknown function v appears on both sides. This equation arises as follows: If the initial state is θ_0 and the decision maker employs σ , then he takes action $\sigma(\theta_0)$ initially, giving immediate reward $r(\theta_0, \sigma(\theta_0))$. Moreover, the state at time $t = 1$ is θ with probability $\pi(\theta | \theta_0, \sigma(\theta_0))$. Starting next period, if the state is θ , the discounted sum of all rewards from next period on, discounted back to $t = 1$, is just $v(\theta, \sigma)$; this is so because σ is stationary. So the discounted expected value of all future rewards from using σ , discounted back to the present time $t = 0$, is $\gamma \sum_{\theta} v(\theta, \sigma) \pi(\theta | \theta_0, \sigma(\theta_0))$. The recursive equation, then, says that the value of following σ is the sum of the current reward and the discounted sum of all future rewards. With this recursive equation and a bit of luck or skill, we may be able to find $v(\theta, \sigma)$ (as we did in the example).

Having this recursive solution and being able to solve it are two different things, of course. Moreover, we can't be sure that every solution of the recursive equation is, in fact, the value function v . For example, if r is bounded, then $v \equiv +\infty$ and $v \equiv -\infty$ are both solutions of this equation. In the theory of dynamic programming, characterizations of solutions are given for particular special cases. A sample of typical results is

Proposition 1. (a) Suppose r is a bounded function. Then $v(\cdot, \sigma)$ is the unique bounded solution of the recursive equation.^e

(b) If r is bounded below, then $v(\cdot, \sigma)$ is the smallest solution of the recursive equation that is bounded below. If r is bounded above, then $v(\cdot, \sigma)$ is the greatest solution of the recursive equation that is bounded above.

(c) Suppose the range of possible states is a finite set Θ . Let Π be the $\Theta \times \Theta$ matrix whose θ, θ' term is $\pi(\theta' | \theta, \sigma(\theta))$. Then $v(\cdot, \sigma)$ is the vector $r(\sigma(\cdot), \cdot) \cdot (I - \gamma \Pi)^{-1}$, where I is the appropriately sized identity matrix. (Part of this statement is the assertion that the matrix $(I - \gamma \Pi)$ is in fact invertible.)^f

The optimal value function, optimal strategies, and Bellman's equation

For every conceivable strategy σ , stationary or not, we can compute $v(\cdot, \sigma)$, the expected value of using σ , considered as a function of the initial state. The *optimal value function* is then defined as

$$f(\theta) = \sup_{\sigma} v(\theta, \sigma),$$

^e We remind the careful reader that $\gamma < 1$ is assumed throughout. In treatments of dynamic programming where this restriction on γ is removed (a) would not hold.

^f If the notion of the inverse of an infinite dimensional matrix is not too much for you to stomach, part (c) generalizes to countable state spaces as long as r is bounded.

where the supremum is taken over all conceivable strategies σ . And strategy σ is optimal if $f(\theta) = v(\sigma, \theta)$ for all θ . Moreover, we have the following recursive equation for f .

Proposition 2. The optimal value function f satisfies the recursive equation

$$f(\theta_0) = \sup_{\alpha \in A(\theta_0)} \left[r(\theta_0, \alpha) + \gamma \sum_{\theta} f(\theta) \pi(\theta | \theta_0, \alpha) \right].$$

This is known as *Bellman's equation* or the *optimality equation*. The idea it expresses is that the optimal value is obtained if one takes the best possible action in the first period and then employs an optimal strategy. It takes a proof (and, in fact, it requires some technical conditions, conditions similar to those that guarantee all the expectations that make up value functions are well-defined). But it should not be so hard to believe.

Conserving strategies

Suppose we could find a strategy σ that attained the supremum in Bellman's equation for every state θ . That is, for all θ_0 , σ satisfies

$$f(\theta_0) = r(\theta_0, \sigma(\theta_0)) + \gamma \sum_{\theta} f(\theta) \pi(\theta | \theta_0, \sigma(\theta_0)).$$

Such a strategy is called *conserving*, because it conserves the optimal value; if you use this strategy for one stage and then follow an optimal strategy (which gives you value f), you will get the optimal value.

Proposition 3. (a) Any optimal strategy σ is conserving.

(b) If r is bounded above, then any conserving strategy is optimal.

In other words, being a conserving strategy is equivalent to being optimal for a wide range of problems, including all those where r is bounded.

The qualification is necessary, as the following simple example demonstrates: Suppose $\Theta = \{0, 1, 2, \dots\}$, $A(\theta) = \{\theta + 1, 0\}$ for $\theta > 0$, and $A(0) = \{0\}$. Transitions from one state to the next are very simple; you go with certainty to the state that has the same label as the action you select. That is, from state $\theta > 0$ you can choose to go either to $\theta + 1$ or to 0. Once you get to state 0, you never get out. Rewards are given by

$$r(\theta, \alpha) = \begin{cases} 0 & \text{if } \alpha = \theta + 1, \text{ and} \\ (1/\gamma)^\theta & \text{if } \alpha = 0, \end{cases}$$

if $\theta > 0$, and $r(0, 0) = 0$. That is, if you choose to make a transition to state 0 from $\theta > 0$, you receive a reward whose size grows with θ in a way that just offsets the discount rate. In all other cases, you get zero reward. You can show that in this case $f(n) = 1/\gamma^n$; if you start at $t = 0$ in state $\theta = n$, then as long as you make the transition to state 0 sometime, you get $1/\gamma^n$ (in present value terms). But then one conserving strategy is $\sigma(\theta) = \theta + 1$; always put off making the transition. It is conserving, because there is never any hurry to make the transition to 0, as long as you do it. The problem is that with this particular conserving strategy you never make the transition to 0, so you never accrue any reward.

Once you have digested this example, consider a simple variation in which the reward of moving from θ to 0 is $(1/\gamma)^\theta - 1$. In this case, things get better and better the longer you delay the transition to state 0. But if you never make the transition, you never get a positive reward. What is f in this case? Which strategies are conserving? Which strategies are optimal?

The intuition behind proposition 3(b) can now be seen. If r is bounded above, then the value of rewards accrued in the distant future must vanish (because of the discount factor). So "conserving" the optimal value does not mean putting off an ever bigger prize, and (one can prove) conserving strategies are optimal.

In what ways is the notion of a conserving strategy useful? To apply this notion, you have to know the optimal value function f . In most problems, you will not know what f is, and so as a means of finding a specific optimal strategy, this concept isn't useful. (In a few cases you can determine what f is from first principles. And you can sometimes "compute" f by approximating an infinite horizon problem with infinite horizon problems; this is called *value iteration* in the literature.)

But there is another way to use the notion of a conserving strategy. Suppose you can argue from first principles and assumptions made that a conserving strategy exists. For example, if each $A(\theta)$ is finite, then there must be a conserving strategy, since then the supremum is automatically a maximum. Then if r is bounded above, you know that an optimal strategy exists, which is not otherwise guaranteed.⁹

Sometimes you can argue from first principles that f takes a particular form. And then you can get a lead on what an optimal strategy must look like. A particular example of this is our consumption-savings problem. Recall the argument we employed to show that, for our candidate strategy σ , $v(\cdot, \sigma)$ takes the form Ky^5/p^{25} for some constant K . By a similar argument it can be shown that the optimal value function f takes the form Ly^5/p^{25} for some L . It may be helpful to see a bit of the argument sketched. (You are asked to give the complete argument in problem 5.) Take any y and p and any strategy σ that is

⁹ For example, see the variation sketched three paragraphs ago.

feasible starting at (y, p) . Consider starting the problem with λy and p for $\lambda > 0$, and consider modifying σ so that the consumer simply scales up (or down) all consumption and savings decisions by the constant λ . This strategy will be feasible starting at $(\lambda y, p)$. And it will deliver utility $\lambda^{-5}v(\sigma, (y, p))$. Hence $f((\lambda y, p)) \geq \lambda^{-5}v(\sigma, (y, p))$ for all σ . Since σ is arbitrary here, $f((\lambda y, p)) \geq \lambda^{-5}f((y, p))$. And since λ can be replaced by $1/\lambda$ and y by λy in this inequality, we see that $f((\lambda y, p)) = \lambda^{-5}f((y, p))$. By a similar argument, $f((y, \lambda p)) = f((y, p))/\lambda^{25}$. If you work at it, you will see that these two equations establish that f takes the form we indicated. (Hint: Let $L = f(1, 1)$.)

Knowing that f takes the form Ly^5/p^{25} is extremely helpful in one way. Once we know this, we can work out the form of any conserving strategy from Bellman's equation. It should come as no surprise to learn that if f takes this form, then conserving strategies must be of the form of our candidate solution. Spend a constant fraction of y on current consumption and save the rest, and divide current spending half each on artichoke and broccoli. We don't know, of course, the constant k of y that is spent on current consumption until we know what L is. But we know that the optimal strategy, since it is conserving, must take this form. And then we can proceed to the next bit of technique.^h

Unimprovable strategies

The strategy σ is said to be *unimprovable* if it satisfies Bellman's equation with $v(\cdot, \sigma)$ in place of f ; i.e., if for all θ_0 ,

$$v(\theta_0, \sigma) = \sup_{\alpha \in A(\theta_0)} \left[r(\theta_0, \alpha) + \gamma \sum_{\theta} v(\theta, \sigma) \pi(\theta | \theta_0, \alpha) \right],$$

or, equivalently, if $\sigma(\theta_0)$ achieves the supremum in the expression on the right-hand side for every θ_0 . As noted in the previous section, a more apt description of this property is "unimprovability in a single step."

Proposition 4. (a) Any optimal strategy σ is unimprovable.

(b) If r is bounded below, then any unimprovable strategy is optimal.

Here is the canonical example to show why you need to assume that rewards are bounded below. Let $\Theta = \{1, 2, 3, \dots\}$ and let $A(\theta) = \{C, F\}$. No

^h Perhaps you wish to object: In the problem of the previous section, r wasn't bounded above. But this objection doesn't hold water. We are using here the result that every optimal strategy is conserving; once we characterize the nature of all conserving strategies, we a fortiori have characterized the nature of any optimal strategy.

matter what action you take in state θ , transition on the next step is to $\theta + 1$ with probability one. But there is a difference in the two actions: $r(\theta, F) = 0$ while $r(\theta, C) = -(1/\gamma)^\theta$. That is, F is free, and C is costly. Of course, the optimal strategy is always to choose F , yielding $f = 0$. But consider the abysmal strategy σ that always chooses C . This gives a total discounted cost $v(\theta, \sigma) = -\infty$. The point is that σ is unimprovable. If you choose F for one period and then revert back to σ , your expected value will continue to be $-\infty$. The only way to do better than σ is to deviate from it at all but a finite number of periods (and the best thing is to deviate from it all the time); since unimprovability only checks one-step deviations, σ passes the test of unimprovability.

From this example, you may be able to see the intuition behind proposition 4(b). If a strategy is unimprovable in a single step, then (one can show by induction) it is unimprovable if you change it over any finite time horizon. It can be suboptimal, then, only if it is "bad" in the arbitrarily far distant future. But if rewards are bounded below (and we are discounting with a discount factor strictly less than one), no strategy can be that bad in the arbitrarily far distant future. Hence unimprovability over any finite horizon implies optimality.

In the previous section we saw how useful proposition 4 can be.

A2.5. Bibliographic notes and discussion

For the purposes of this book, the functional equation for $v(\cdot, \sigma)$ and proposition 4(b) are what matters. And when we wish to use these two, r will be bounded and Θ will be finite so there will be no difficulties encountered at all.

But the theory of dynamic programming, which is used more and more in economics, goes far beyond the little we've mentioned here. And complementary to the theory of dynamic programming, which envisions decisions made at discrete points in time, is control theory, where the decision maker is continually choosing actions. We recommend the following for further reading: Howard (1960) is an excellent introduction to the subject. Ross (1983) is perhaps one level deeper. Whittle (1982) is authoritative. Stokey and Lucas (1989) is another complete treatment that is especially useful because it is devoted to applications in economic theory.

References

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- Stokey, N.; and Lucas, R. 1989. *Recursive Methods in Economic Dynamics*. Cambridge, Mass.: Harvard University Press.
- Whittle, P. 1982. *Optimization Over Time*, vols. I, II. New York: John Wiley and Sons.

A2.5. Problems

- 1. In the problem of section A2.3, suppose that you can interview up to 4 firms, where each will offer you a wage uniformly distributed on the interval $[0, 10]$. Every time you interview, though, you pay a fee of \$1. What is the optimal strategy to follow if you must take an offer when made, as in section A2.3? What is the optimal strategy to follow if you can always go back to previous offers?
- 2. Now consider problem 1, but in the case where you can interview infinitely many firms at a cost of \$1 per firm.
- 3. In the problem of section A2.3, suppose that if you turn down a job, you can later make (costless) inquiries to see if it is still available, but there is a probability that the job will already have been filled. Because this is a difficult problem, consider the case where there are only three possible jobs. And suppose that if you turn down job 1, there is a .5 probability that it will be filled when you hear the offer made at job 2, and a .75 chance that it will be filled when you hear the offer made at job 3; and if you turn down job 2, there is a .5 probability that it will be filled when you hear the offer made at job 3. What is the optimal strategy to follow? (Hint: Suppose you are offered a wage of 5 at job 1 and 6 at job 2. Do you make inquiries about the availability of the job at firm 1? You may wish to do so, if not for 5 and 6, then for other numbers where job 1 pays less than job 2, and you won't have the answer until you understand why this is.)
- 4. (a) Consider the following problem. You are driving down a long line of parking spaces, looking for a place to park your car. Think of potential parking spaces at locations $-100, -99, -98, \dots, -1, 0, 1, 2, \dots$. You are currently at location -100 , which is unoccupied, and you are considering whether to park there. If you decide not to park there, you will drive a bit further and see whether location -99 is occupied or not; if it is, you must proceed to -98 ; and if not, you must decide whether to park there or try your luck with -98 , and so on. You cannot circle back to empty parking spots; U-turns are strictly forbidden. Each spot is occupied with another car with probability $\alpha < 1$, and whether a spot is occupied is

A2.5. Problems

independent of the occupancy of any and all other slots. You know the value of α . Your objective is to park as close as possible to location 0. More specifically, if you park in location n , then the cost to you is $|n|$, and you seek to minimize the expected cost you incur. You are able to see only the occupancy status of one spot at a time. What should you do? What is the expected cost to you?

(b) Suppose that when you are at spot n , you can see the occupancy status of spot n and the next five spots ahead. What should you do? What is the expected cost to you?

(c) Suppose that you don't know the value of α . Being very specific, suppose $\alpha = .9$ or $\alpha = .7$, with each of these two values being equally likely. (If you know the terminology, this makes the occupancy status of the spots exchangeable instead of being independent.) What should you do? What is the expected cost to you? (Use the formulation where you cannot see ahead. A numerically derived solution is quite adequate. An important first step is to decide on what links the "past" to the "future" in this problem.

■ 5. (a) Give a rigorous proof of the assertion that if in the consumption savings problem of section A2.3 you spend constant and equal fractions of your wealth on artichoke and broccoli (and save the rest), then your discounted expected utility takes the form $Ky^{.5}/p^{.25}$, where y is your current wealth and p is the current price of broccoli.

(b) Give a rigorous proof of the assertion that the optimal value function f in this context takes the form $Ly^{.5}/p^{.25}$.

■ 6. Suppose in the consumption-savings problem of section A2.3 the price of broccoli is constant (and equal, say, to 2), but now there are two ways to save. You can put your savings in a bank, where \$1 put in today returns \$1.02 tomorrow. Or you can invest in a risky asset that for every \$1 invested on one day returns either \$1.07 or \$.99 the following day, with each of these values equally likely. (The return on the risky asset is independent from one day to the next.) What is the optimal strategy to follow in this case?

■ 7. Consider the variation on the example given following proposition 3. Are there any conserving strategies in this case? Are there any optimal strategies?

addendum to the second and subsequent printings

A few changes have been made for the second and subsequent printings of this book. Most importantly, the index has been revised. Some minor typos and blemishes in the first printing have also been corrected; none are so substantial that confusion should result from the use of copies from the first printing together with copies from subsequent printings. I am grateful to Dr. Ben J. Heijdra and Prof. Tzong-rong Tsai for providing me with lists of typographical errors.

I am aware of several errors in the book which I have decided not to correct at this time; I believe that the changes required would be sufficient to cause confusion from the use of copies from the first printing together with copies from subsequent printings. But I note those errors here, so readers will not be led too far astray. (I suggest that you mark in the margins of appropriate pages a note to see this page, or alternatively you make the changes manually. Please note that owners of copies from the first printing will be unaware of these errors; you might wish to share these changes with them.) Thanks to Prof. Andrew Weiss for alerting me to the problems with chapter 17 discussed following.

On page 261, the last part of problem 7.10 is obscure at best and wrong under the most reasonable interpretation of the problem I can think of. Instructors should be careful assigning this problem.

Also page 261, add the following assumptions to problem 7.11: $F : R_+^N \rightarrow R_+$, $\phi : R_+ \rightarrow R_+$, $\phi(0) = 0$, $\lim_{a \rightarrow \infty} \phi(a) = \infty$, and ϕ is continuous (in addition to being strictly increasing). In view of the error in problem 7.10 (see just above), part (c) of this problem is silly.

On page 269, proposition 8.1 needs further assumptions. Assume (a) the cost of at least one variable factor of production is strictly positive, and either (b1) $f(\hat{x}_1, \dots, \hat{x}_{N'}, 0, \dots, 0) = 0$ where 1 to N' index the fixed factors of production, or (b2) restrict attention to output quantity levels for which the fixed factors of production are insufficient. Note that these further assumptions will be needed in answering problem 8.1 (page 292).

On page 436, the *tests of dominated messages* and *equilibrium domination* are flawed. Let me give a correct version of the test of equilibrium domination. An equilibrium (π_1, π_2) fails the test if, for some t' and s' that is

equilibrium dominated at t' , and for every $\pi'_2(\cdot, s')$ that is a conditional best response to s' given beliefs that put zero probability on s' being sent by t' , $\sum_{a \in A} u(t, s', a) \pi'_2(a, s') > u^*(t)$ for some t . (Indeed, one can sharpen this as follows: For given s' , let $T(s')$ be the set of t such that s' is equilibrium dominated at t . The equilibrium fails the test if there exists some subset T' of $T(s')$ with nonempty T/T' , such that for every $\pi'_2(\cdot, s')$ that is a conditional best response to s' given beliefs that put zero probability on T' given s' , there is some t such that $\sum_{a \in A} u(t, s', a) \pi'_2(a, s') > u^*(t)$.) The careful reader will note that on page 436, the quantifiers are mangled; if you know something about the subject, you will see that my screw-up is intimately related to why Kohlberg and Mertens are forced to move to a set-valued solution concept (cf. page 443 of the text). Put another way, the tests as given on page 436 are too severe; in cases of interest, no single equilibrium will pass these tests. Since the cryptic correction given here is apt to be hard to parse, the confused reader should (a) disregard the precise formulations of the tests given on page 436 and (b) go back to Cho and Kreps (1987) for (I hope) a more satisfactory exposition.

On page 451, problem 12.1, to be very precise, you must assume that the game tree has at least one node.

In chapter 17, difficulties can arise because (on page 630) I assume that the range of possible education signals is bounded above. Subsequently (for example, in the proof of proposition 3 on page 640) I implicitly assume that education levels can always be raised. Put another way, we could have a pool at the upper limit of possible education levels. I believe the easiest fix is to assume that the range of possible education signals is unbounded, or assume that the cost of obtaining the highest possible level of education (at any possible wage level) is prohibitive relative to the zero-education, zero-wage outcome.

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